

Project Overview

Project Details



Intern Name: Muhammad Talha

Internship Program: Cyberster Blue Team Internship

Organization: Cyberster

Week 4 (Days 22–28)

Date: July 16, 2026

Status: Completed

Introduction

During Weeks 1–3 of the Cyberster Blue Team Internship, the focus was on establishing a foundational Security Information and Event Management (SIEM) environment using **Wazuh**. The environment included endpoint monitoring, File Integrity Monitoring (FIM), and network intrusion detection through **Suricata IDS**. These

components provided visibility into host-level and network-level activities, forming the basis for proactive threat detection.

In Week 4, the internship progressed toward building a **complete defensive infrastructure** by integrating a **pfSense Firewall** as the network perimeter device. This week's objectives included deploying pfSense, routing all lab traffic through it, configuring firewall policies, forwarding logs to Wazuh, integrating external threat intelligence sources such as **VirusTotal**, mapping detections to the **MITRE ATT&CK** framework, enabling **vulnerability detection**, and developing a **custom SOC dashboard** for centralized monitoring.

This phase marked the transition from isolated host monitoring to a fully integrated **Security Operations Center (SOC) environment**, capable of correlating network, endpoint, and threat intelligence data for comprehensive defense.

Weekly Objectives

-
- Deploy pfSense as the perimeter firewall
- Configure WAN and LAN interfaces
- Route laboratory traffic through pfSense
- Configure firewall policies and enable logging
- Forward pfSense logs to Wazuh Manager
- Integrate external Threat Intelligence (VirusTotal API)
- Explore MITRE ATT&CK mapping within Wazuh
- Enable Vulnerability Detection module
- Create a customized SOC monitoring dashboard
- Produce professional security reports

Lab Environment

The Week 4 lab environment consisted of the following virtualized components:

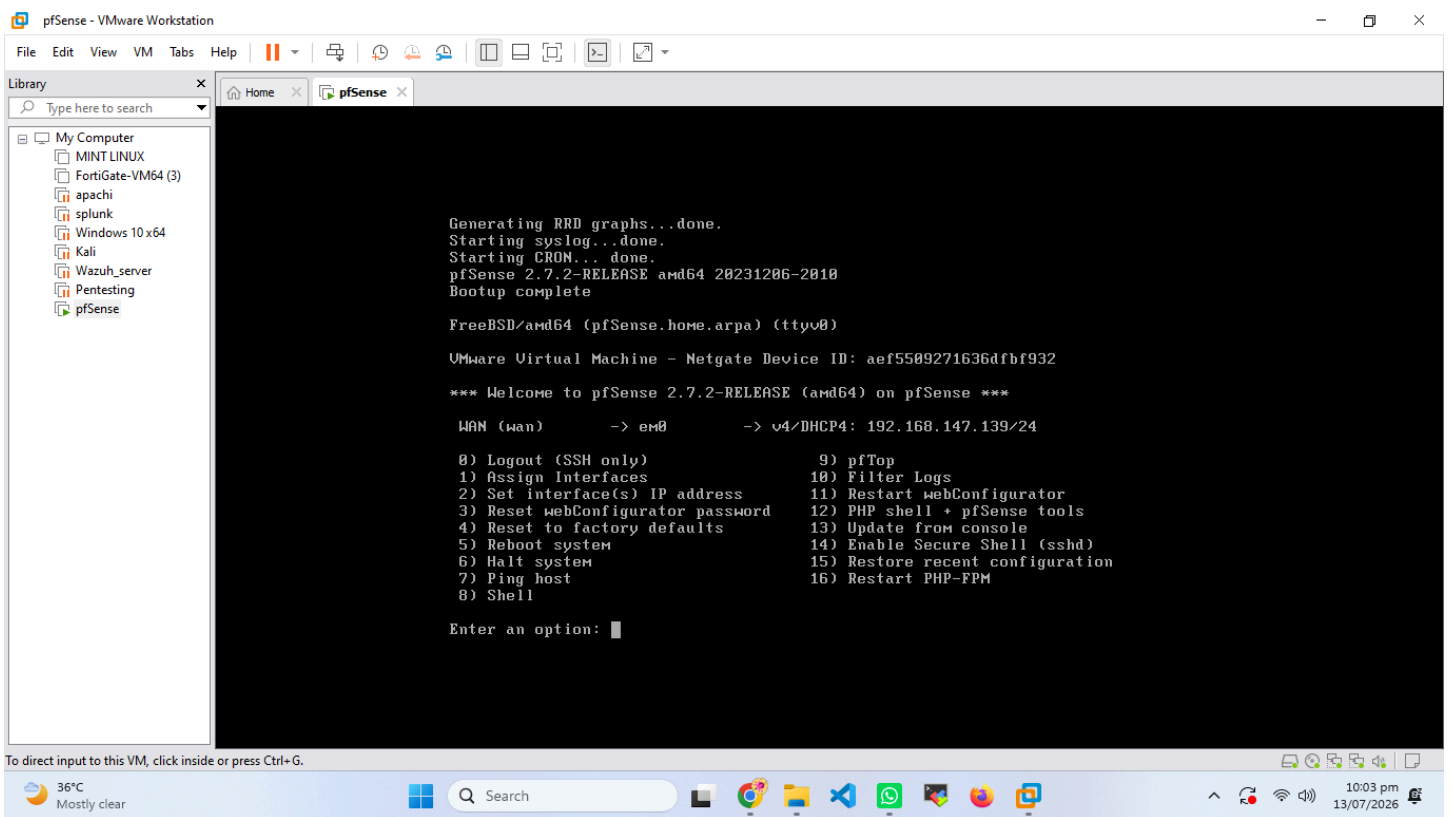
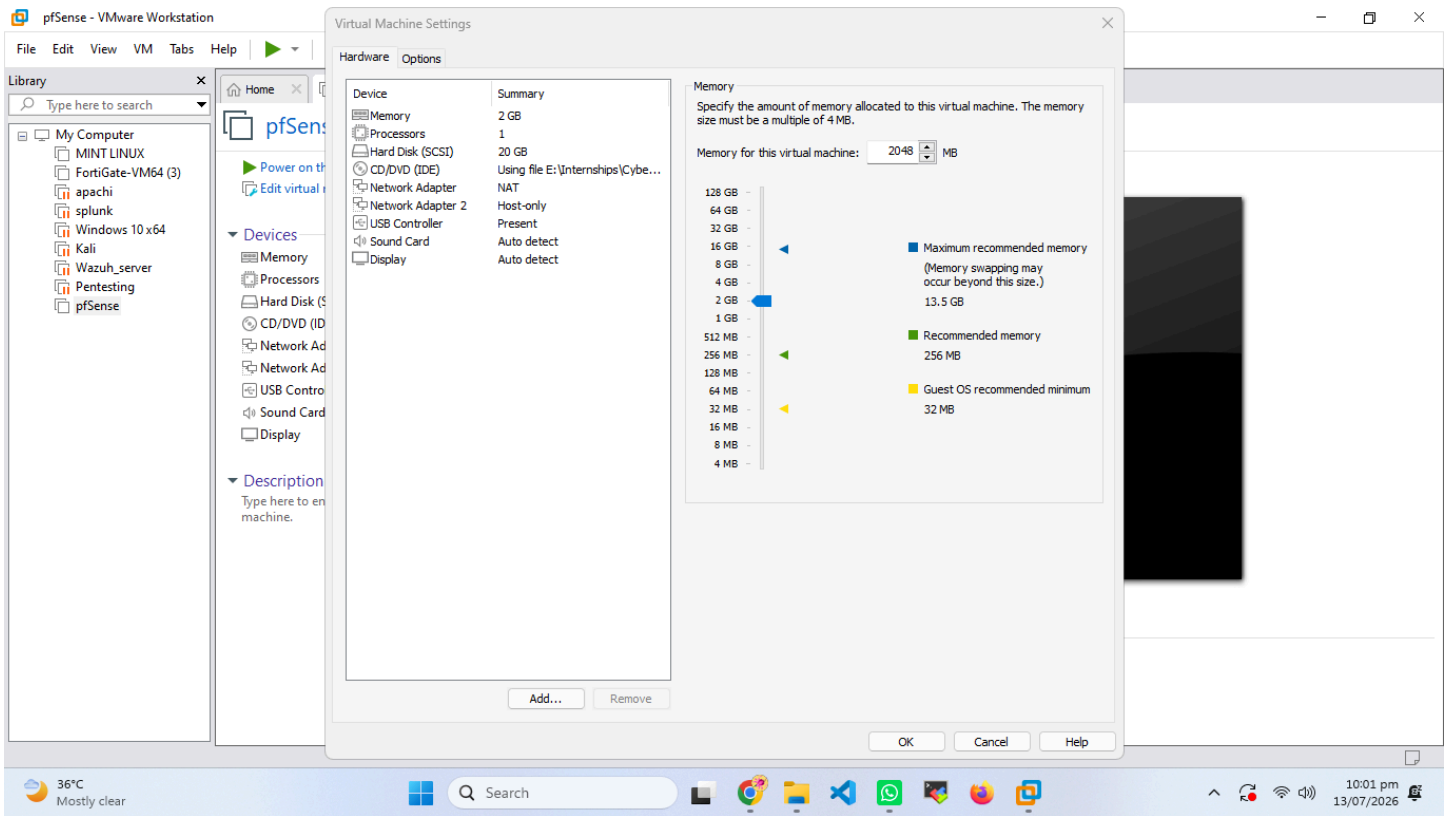
System	Role	Operating System	Purpose
Ubuntu Server	Wazuh Manager	Ubuntu 22.04 LTS	Centralized SIEM and log management
Windows 10	Monitored Agent	Windows 10 Pro	Endpoint monitoring and event generation
Kali Linux	Attack Simulation	Kali Linux 2024	Penetration testing and threat emulation
pfSense	Firewall & Router	pfSense CE 2.7	Network perimeter security and routing
VMware Workstation	Virtualization Platform	Host System	Virtual network and system management

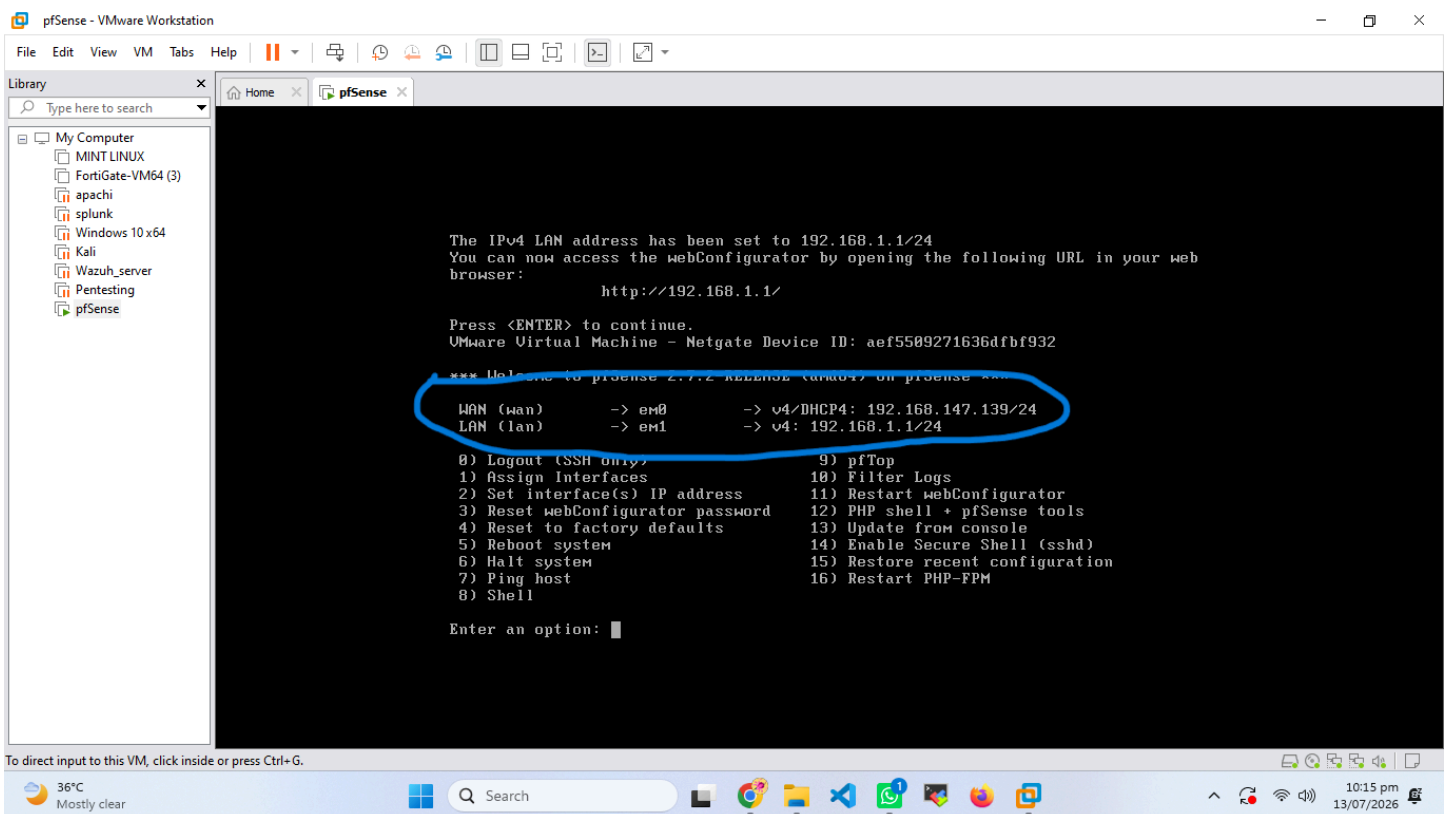
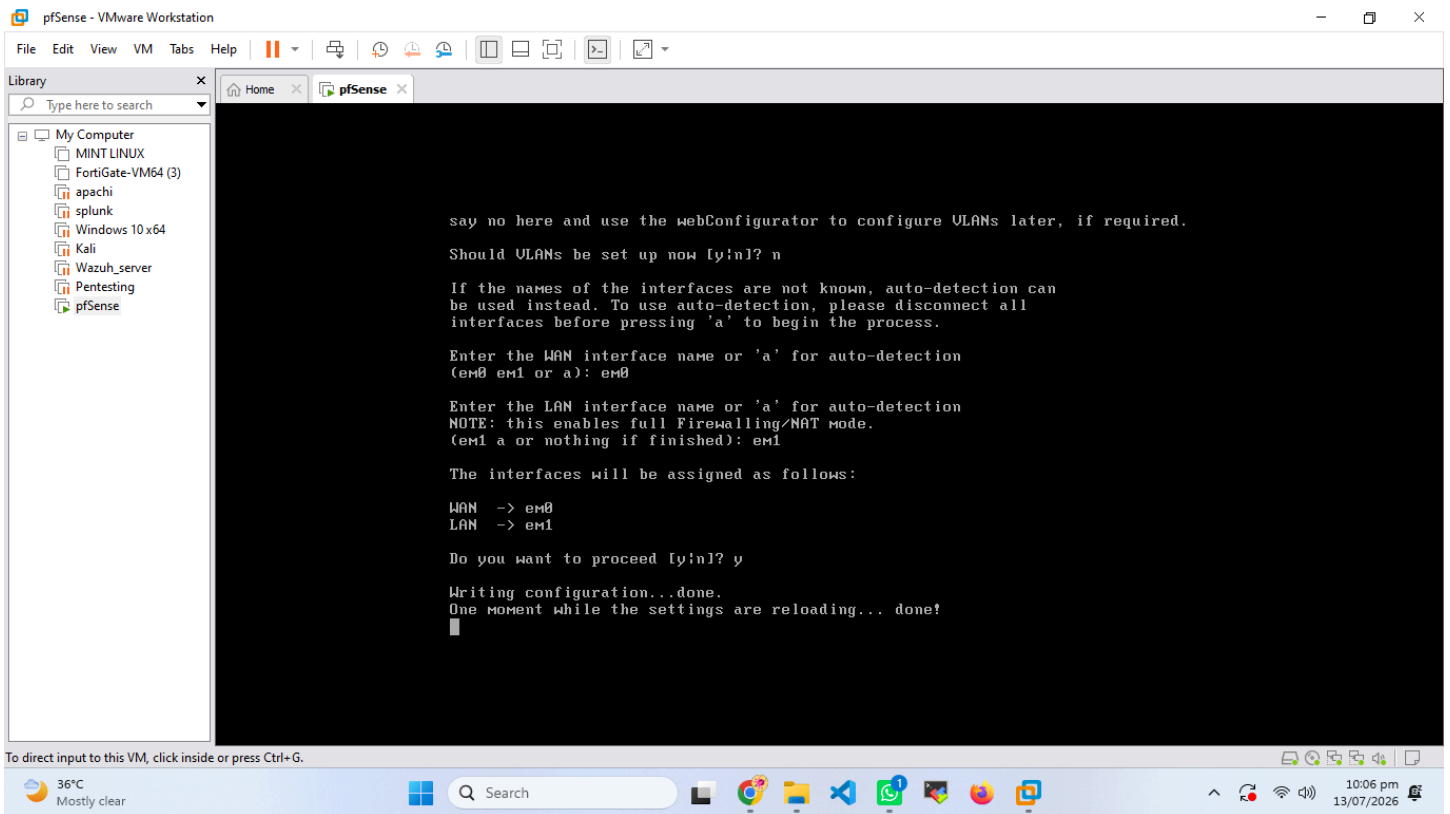
Existing Network Architecture

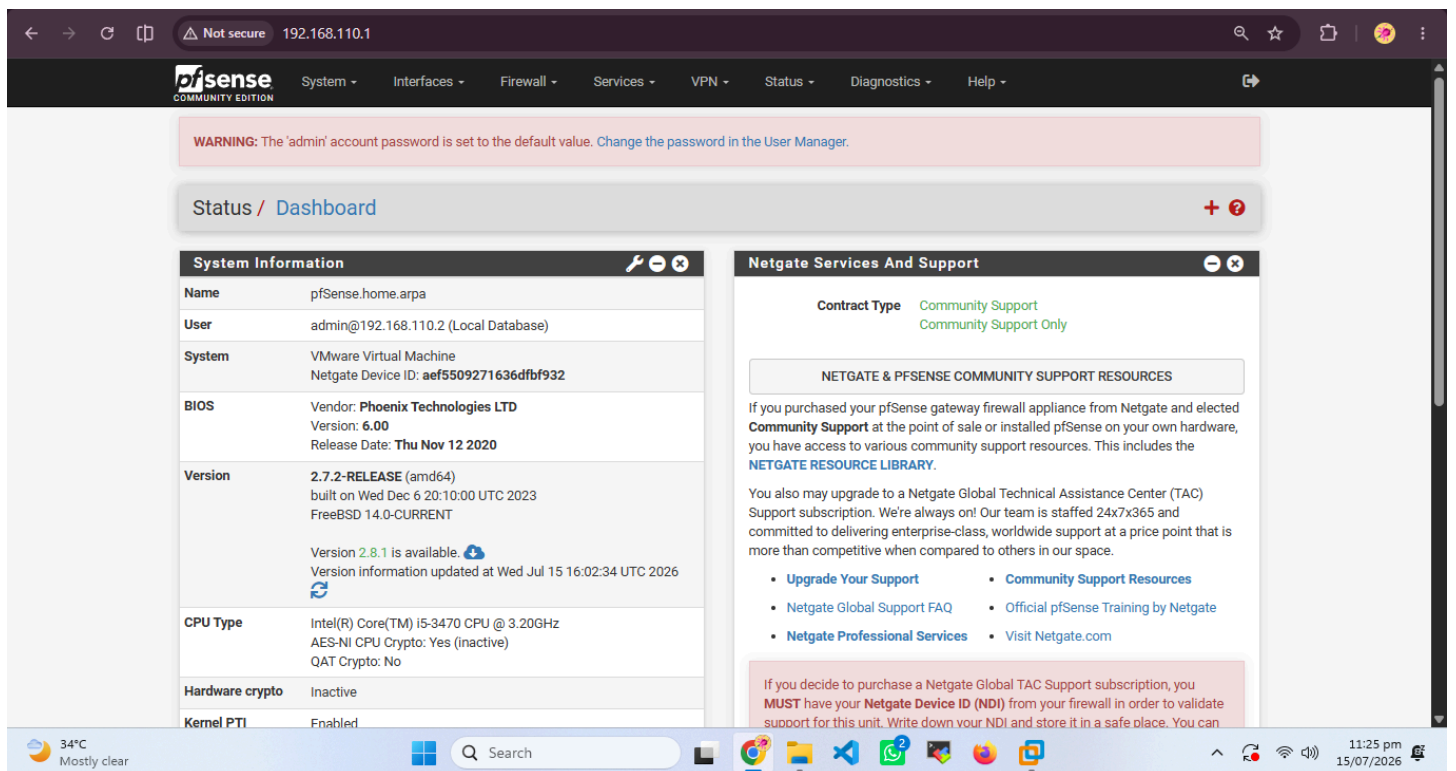
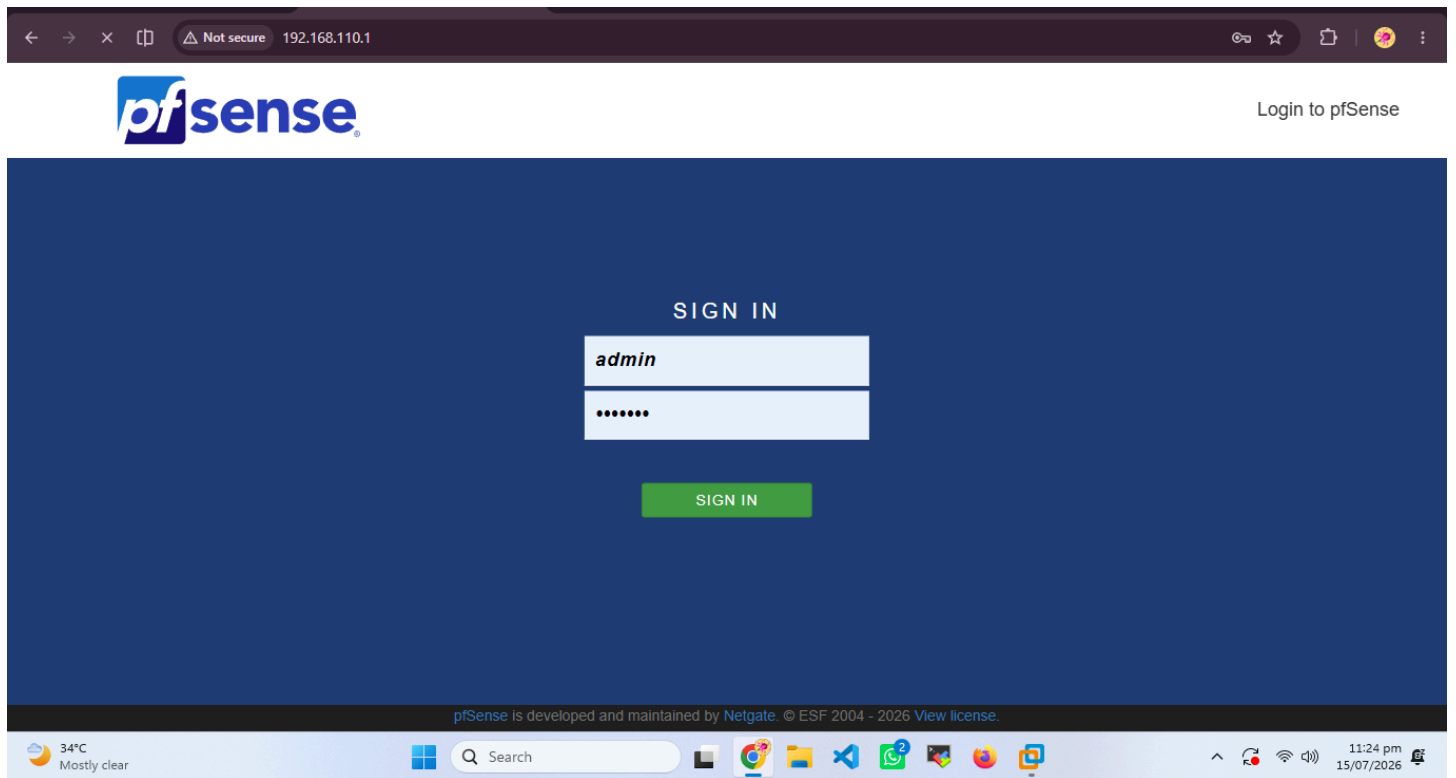
Initially, the lab operated on a **Host-Only network** where all virtual machines communicated directly without a dedicated gateway. This setup limited visibility into network traffic and lacked centralized control.

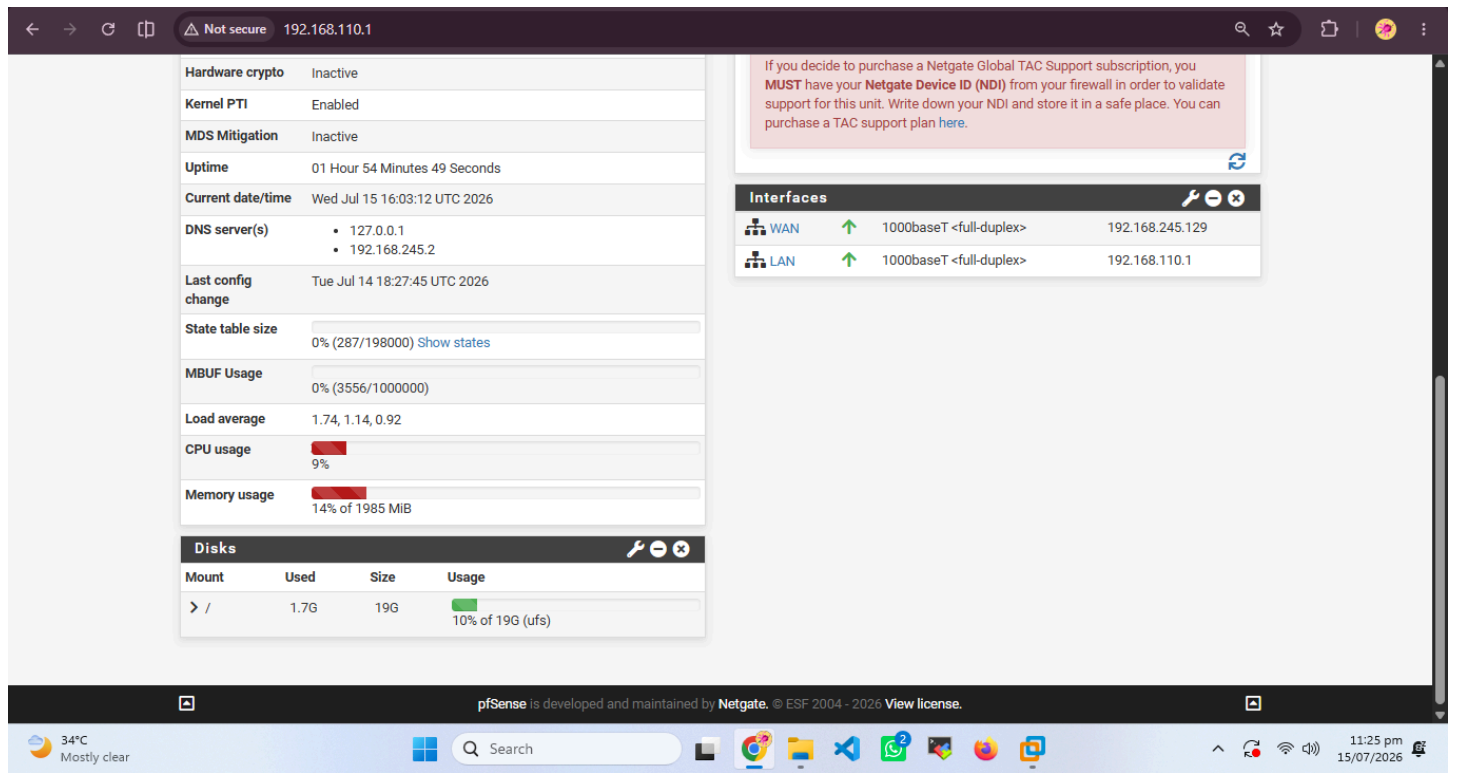
In Week 4, the architecture transitioned to a **pfSense-based gateway model**, introducing a structured network topology with **WAN** and **LAN** interfaces. All internal traffic was routed through pfSense, enabling:

- Centralized traffic monitoring and control
- Firewall policy enforcement
- Comprehensive traffic logging
- Network segmentation between internal and external zones
- Enhanced visibility for SOC analysis
- A more realistic enterprise-style network environment









pfSense Firewall Deployment

The pfSense deployment began with creating a new virtual machine in VMware Workstation using the pfSense ISO image. Two network adapters were configured:

- **Adapter 1 (WAN):** Connected to NAT for Internet access
- **Adapter 2 (LAN):** Connected to Host-Only network for internal communication

After installation, the pfSense console menu was verified to ensure both interfaces were detected correctly. The web interface was accessed via the default LAN IP (192.168.1.1) using a browser on the Windows 10 system. The initial setup wizard was completed, setting the hostname, domain, DNS servers, and admin credentials.

← → ↺ ↻

Not secure 192.168.110.1/firewall_rules.php?if=lan

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pfSense

COMMUNITY EDITION

System ▾

Interfaces ▾

Firewall ▾

Services ▾

VPN ▾

Status ▾

Diagnostics ▾

Help ▾

↺

WARNING: The 'admin' account password is set to the default value. [Change the password in the User Manager.](#)

Firewall / Rules / LAN

📊 📋 ?

The firewall rule configuration has been changed.
The changes must be applied for them to take effect.

✓ Apply Changes

Floating

WAN

LAN

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	✓ 2/1.94 MiB	*	*	*	LAN Address	80	*	*		Anti-Lockout Rule	⚙️
☐	✓ 0/3.37 MiB	IPv4 *	LAN subnets	*	*	*	*	none		Default allow LAN to any rule	🔗 🖋️ 📄 🗑️ ✖️
☐	✓ 0/0 B	IPv6 *	LAN subnets	*	*	*	*	none		Default allow LAN IPv6 to any rule	🔗 🖋️ 📄 🗑️ ✖️

⬆ Add

⬇ Add

🗑 Delete

🔄 Toggle

📄 Copy

💾 Save

⬆ ⬇ Separator

📘

33°C Mostly clear

🔍 Search

🏠

🌐

📁

🔗

📞

🎮

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📺

⬆ ⬇

🔄

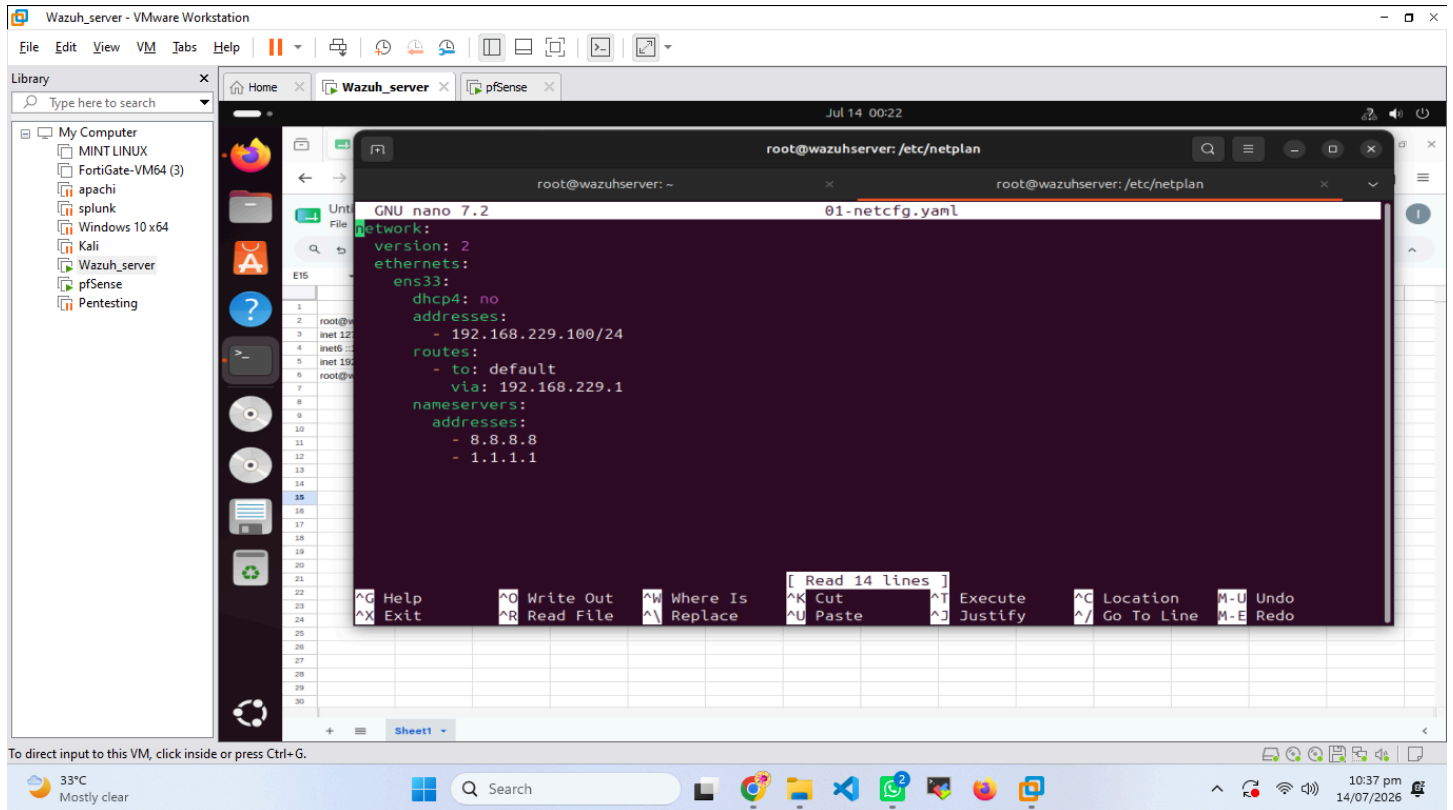
📶

🔊

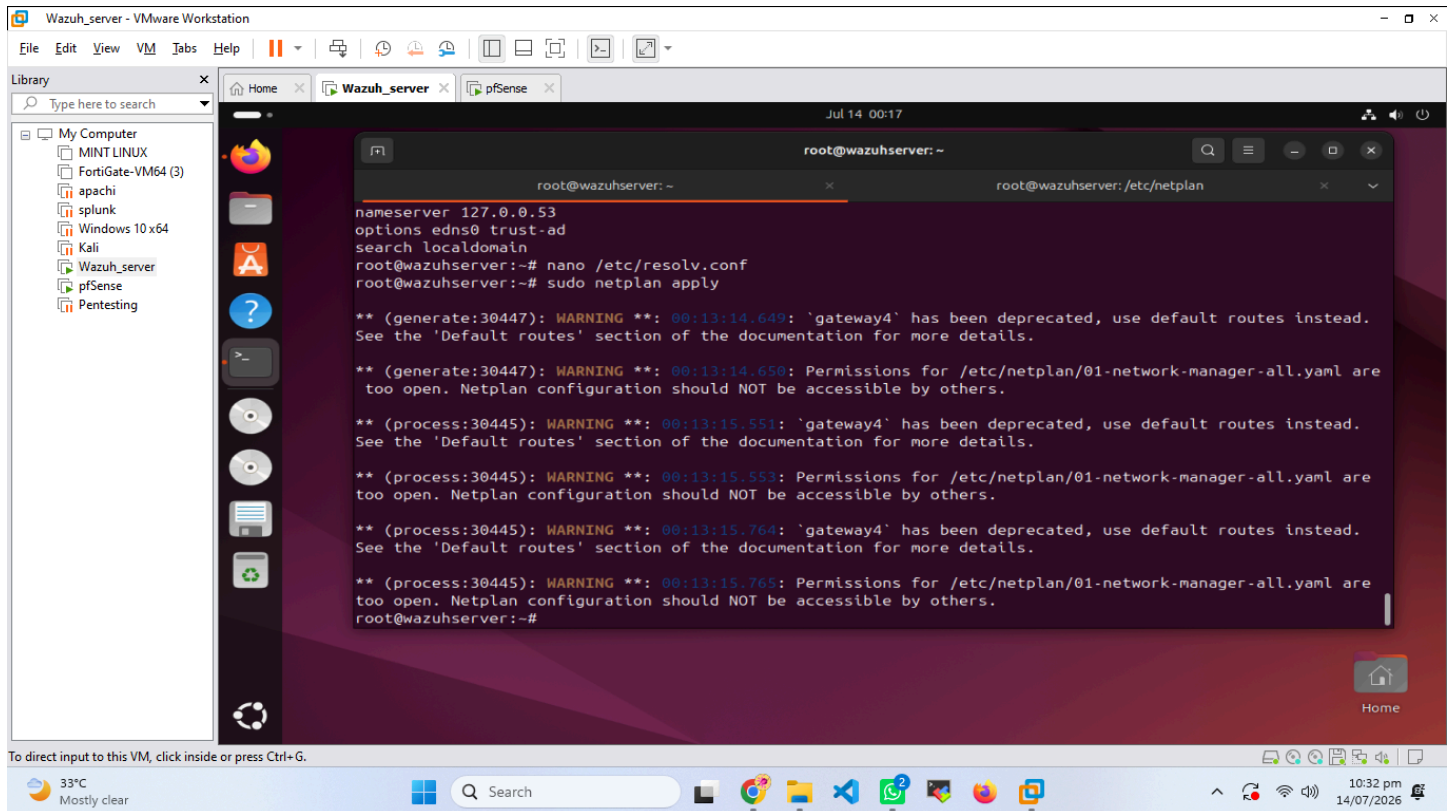
11:41 pm 15/07/2026

- **WAN Interface:** Configured to use DHCP via the NAT adapter, automatically obtaining an external IP address.
- **LAN Interface:** Assigned a static IP (192.168.1.1/24) for internal devices.

- **Default Gateway:** Set to pfSense LAN IP for all internal systems.
- **Connectivity Testing:** Verified using ICMP ping from Windows and Kali systems to pfSense and external Internet hosts. Successful responses confirmed proper routing and NAT functionality.



```
root@wazuhserver: /etc/netplan
GNU nano 7.2 01-netcfg.yaml
network:
  version: 2
  ethernet:
    ens33:
      dhcp4: no
      addresses:
        - 192.168.229.100/24
      routes:
        - to: default
          via: 192.168.229.1
      nameservers:
        addresses:
          - 8.8.8.8
          - 1.1.1.1
```



```
root@wazuhserver: ~
nameserver 127.0.0.53
options edns0 trust-ad
search localdomain
root@wazuhserver:~# nano /etc/resolv.conf
root@wazuhserver:~# sudo netplan apply

** (generate:30447): WARNING **: 00:13:14.649: 'gateway4' has been deprecated, use default routes instead.
See the 'Default routes' section of the documentation for more details.

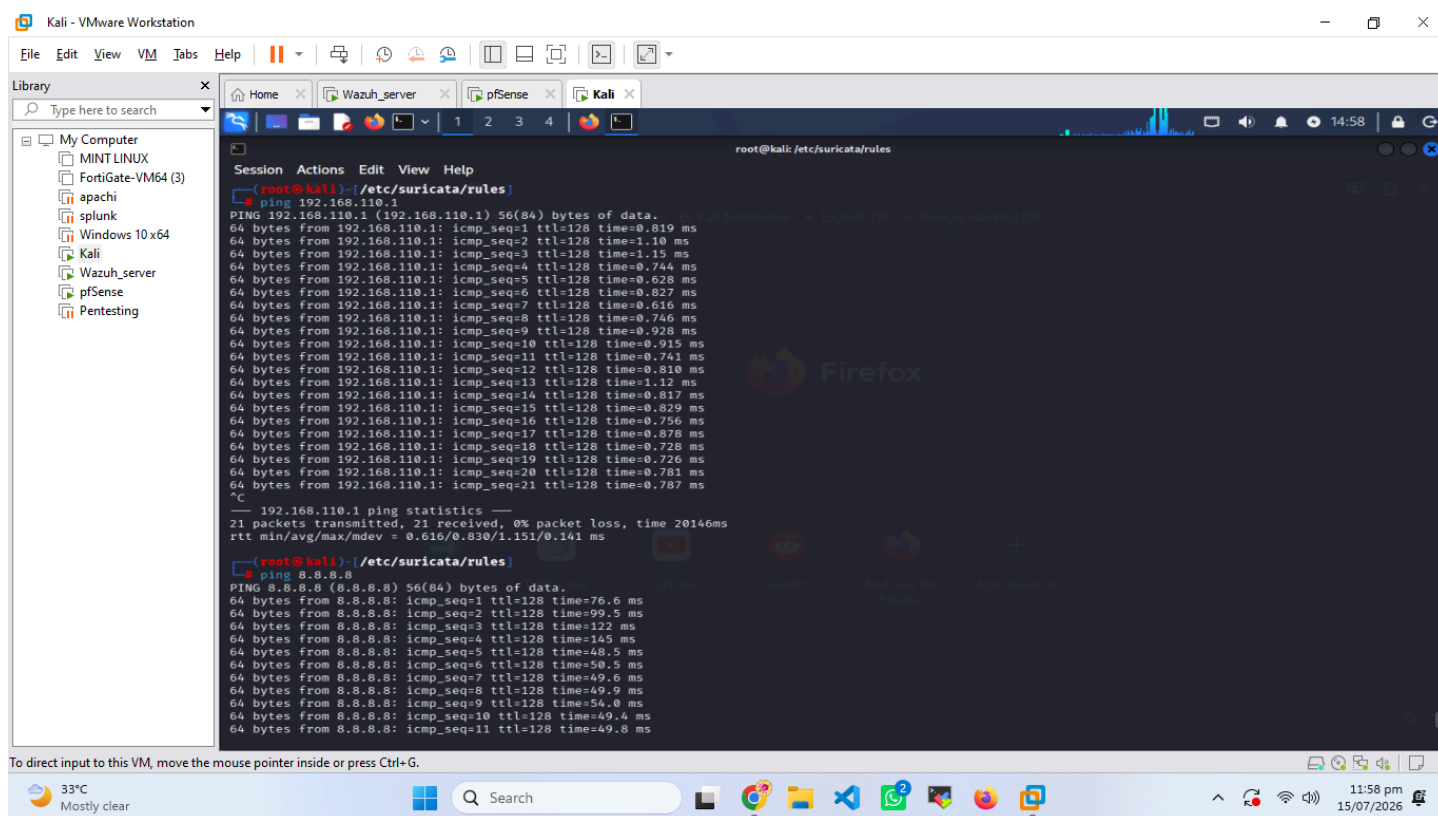
** (generate:30447): WARNING **: 00:13:14.650: Permissions for /etc/netplan/01-network-manager-all.yaml are
too open. Netplan configuration should NOT be accessible by others.

** (process:30445): WARNING **: 00:13:15.551: 'gateway4' has been deprecated, use default routes instead.
See the 'Default routes' section of the documentation for more details.

** (process:30445): WARNING **: 00:13:15.553: Permissions for /etc/netplan/01-network-manager-all.yaml are
too open. Netplan configuration should NOT be accessible by others.

** (process:30445): WARNING **: 00:13:15.764: 'gateway4' has been deprecated, use default routes instead.
See the 'Default routes' section of the documentation for more details.

** (process:30445): WARNING **: 00:13:15.765: Permissions for /etc/netplan/01-network-manager-all.yaml are
too open. Netplan configuration should NOT be accessible by others.
root@wazuhserver:~#
```



Firewall Rule Configuration

pfSense's default configuration allows outbound traffic from LAN to WAN while blocking unsolicited inbound connections. Custom rules were added to:

- Allow ICMP, HTTP, and HTTPS traffic for testing
- Enable logging for all rules to ensure visibility in Wazuh
- Block unauthorized SSH attempts from Kali Linux to Windows 10

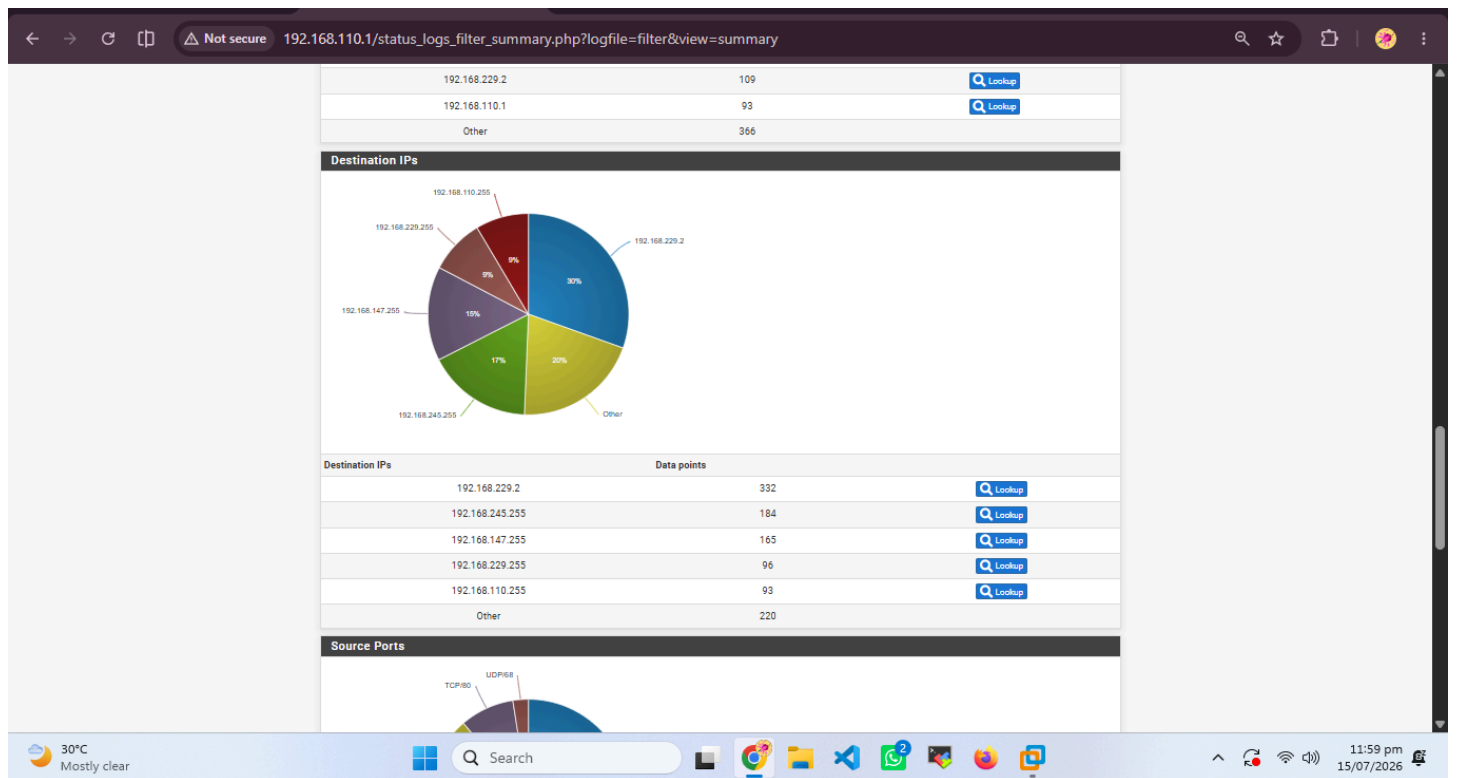
Testing involved initiating connections from Kali to Windows and Ubuntu systems. Logs confirmed that blocked attempts were recorded, validating rule enforcement.

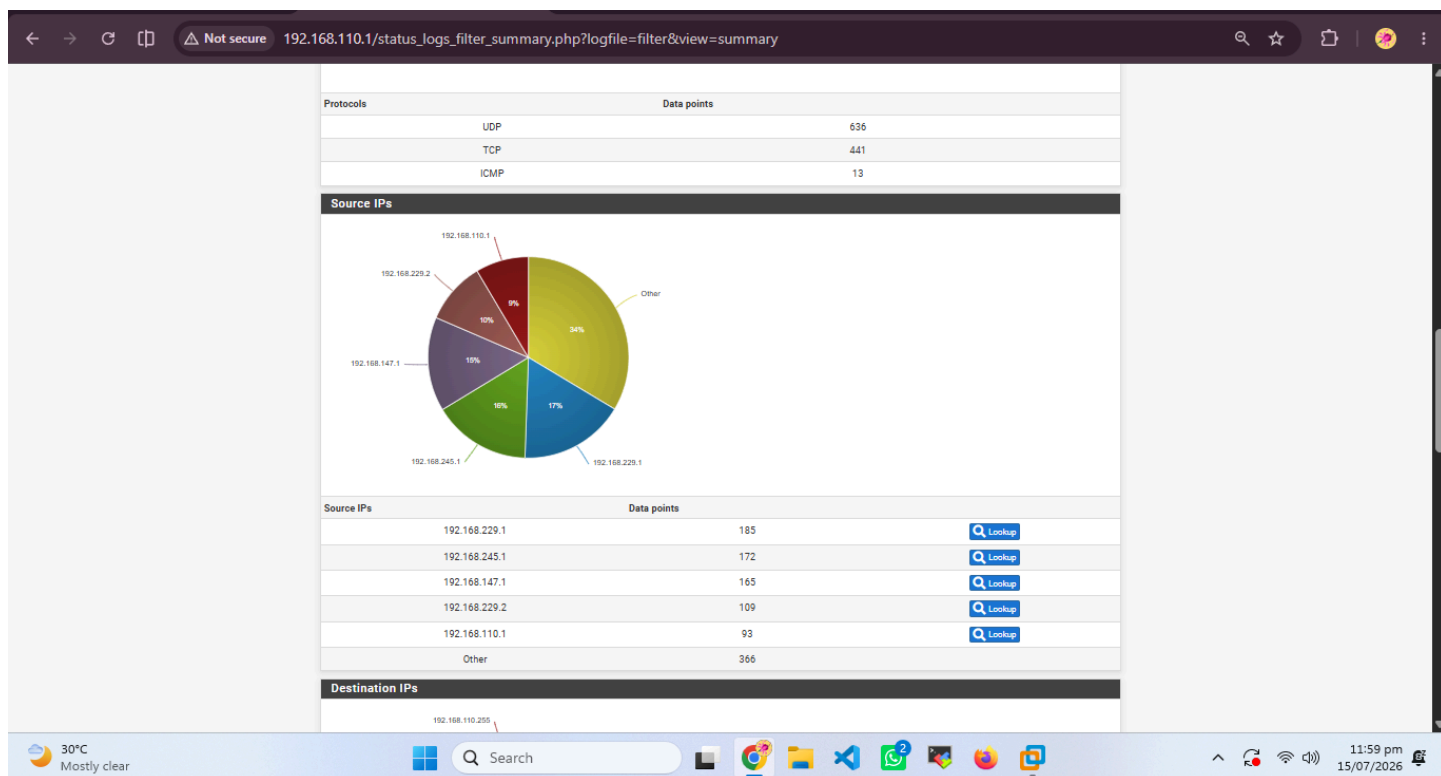
Not secure 192.168.110.1/status_logs_filter.php

✖	Jul 15 18:46:44	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.245.1:137	192.168.245.255:137	UDP
✖	Jul 15 18:46:45	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.245.1:137	192.168.245.255:137	UDP
✖	Jul 15 18:46:46	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.245.1:137	192.168.245.255:137	UDP
✖	Jul 15 18:52:56	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.245.1:137	192.168.245.255:137	UDP
✖	Jul 15 18:52:57	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.245.1:137	192.168.245.255:137	UDP
✖	Jul 15 18:52:58	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.245.1:137	192.168.245.255:137	UDP
✖	Jul 15 18:56:15	WAN	block bogon IPv4 networks from WAN (11001)	0.0.0.0:68	255.255.255.255:67	UDP
✖	Jul 15 18:58:26	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.245.1:137	192.168.245.255:137	UDP
✖	Jul 15 18:58:26	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.245.1:137	192.168.245.255:137	UDP
✖	Jul 15 18:58:27	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.245.1:137	192.168.245.255:137	UDP
✖	Jul 15 18:59:22	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.245.1:137	192.168.245.255:137	UDP
✖	Jul 15 18:59:23	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.245.1:137	192.168.245.255:137	UDP
✖	Jul 15 18:59:24	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.245.1:137	192.168.245.255:137	UDP
✖	Jul 15 18:59:43	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.245.1:137	192.168.245.255:137	UDP
✖	Jul 15 18:59:43	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.245.1:137	192.168.245.255:137	UDP
✖	Jul 15 18:59:44	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.245.1:137	192.168.245.255:137	UDP

Hot days ahead 30°C

12:00 am 16/07/2026





pfSense Logging & Syslog Forwarding to Wazuh

Syslog forwarding was configured under **Status** → **System Logs** → **Settings**. The Wazuh Manager's IP (Ubuntu Server) was added as a remote syslog destination on port 514/UDP.

In Wazuh, a new decoder and rule set were applied to parse pfSense logs, enabling correlation of firewall events with endpoint and IDS alerts. This integration provided unified visibility across network and host layers.

The screenshot displays the pfSense status_logs_settings.php page. It features a warning message, a breadcrumb trail, and a settings form.

WARNING: The 'admin' account password is set to the default value. [Change the password in the User Manager.](#)

Status / System Logs / Settings

The changes have been applied successfully.

General Logging Options

Log Message Format: BSD (RFC 3164, default)
The format of syslog messages written to disk locally and sent to remote syslog servers (if enabled). Changing this value will only affect new log messages.

Forward/Reverse Display: ☐ Show log entries in reverse order (newest entries on top)

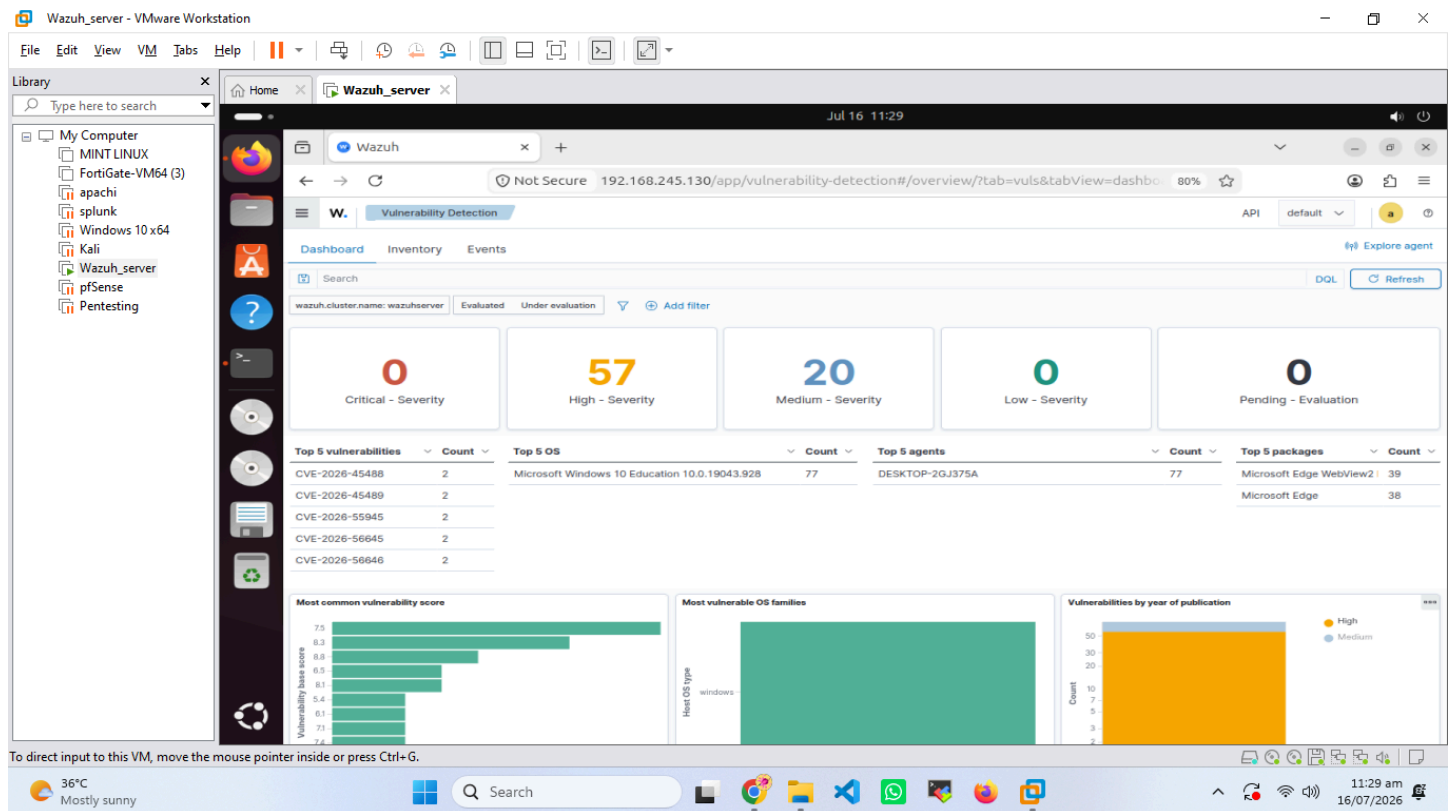
GUI Log Entries: 500
This is only the number of log entries displayed in the GUI. It does not affect how many entries are contained in the actual log files.

Log firewall default blocks: ☒ Log packets matched from the default block rules in the ruleset
Log packets that are **blocked** by the implicit default block rule. - Per-rule logging options are still respected.

Threat Intelligence Integration with VirusTotal

The **VirusTotal API** was integrated into Wazuh to enrich alerts with external threat intelligence. API keys were configured in the Wazuh Manager's `ossec.conf` file. Suspicious file hashes and IP addresses detected by Suricata or endpoint agents were automatically queried against VirusTotal's database.

This integration enhanced detection accuracy by correlating internal alerts with global threat reputation data.



Wazuh_server - VMware Workstation

File Edit View VM Tabs Help

Library

My Computer

- MINT LINUX
- FortiGate-VM64 (3)
- apachi
- splunk
- Windows 10 x64
- Kali
- Wazuh_server
- pfSense
- Pentesting

Wazuh

Jul 16 11:29

Not Secure 192.168.245.130/app/vulnerability-detection#/overview/?tab=vulns&tabView=invento 80%

Vulnerability Detection

Dashboard Inventory Events

Search

wazuh.cluster.name: wazuhservers Evaluated Under evaluation Add filter

77 hits

agent.name	package.name	package.version	vulnerability.description	vulnerability.severity	vulnerability.id
DESKTOP-2GJ375A	Microsoft Edge WebView2 Runtime	148.0.3967.70	Insufficient ui warning of danger...	Medium	CVE-2026-58597
DESKTOP-2GJ375A	Microsoft Edge WebView2 Runtime	148.0.3967.70	Access of resource using incomple...	High	CVE-2026-58295
DESKTOP-2GJ375A	Microsoft Edge WebView2 Runtime	148.0.3967.70	External control of file name or p...	High	CVE-2026-58293
DESKTOP-2GJ375A	Microsoft Edge WebView2 Runtime	148.0.3967.70	Improper input validation in Micr...	High	CVE-2026-58292
DESKTOP-2GJ375A	Microsoft Edge WebView2 Runtime	148.0.3967.70	Operation on a resource after ex...	Medium	CVE-2026-58291
DESKTOP-2GJ375A	Microsoft Edge WebView2 Runtime	148.0.3967.70	Access of resource using incomple...	High	CVE-2026-58290
DESKTOP-2GJ375A	Microsoft Edge WebView2 Runtime	148.0.3967.70	Use after free in Microsoft Edge (...)	High	CVE-2026-58287
DESKTOP-2GJ375A	Microsoft Edge WebView2 Runtime	148.0.3967.70	Use after free in Microsoft Edge (...)	High	CVE-2026-58288
DESKTOP-2GJ375A	Microsoft Edge WebView2 Runtime	148.0.3967.70	Improper access control in Micro...	High	CVE-2026-58286
DESKTOP-2GJ375A	Microsoft Edge WebView2 Runtime	148.0.3967.70	Access of resource using incomple...	High	CVE-2026-58285
DESKTOP-2GJ375A	Microsoft Edge WebView2 Runtime	148.0.3967.70	Improper authorization in Micros...	High	CVE-2026-57983
DESKTOP-2GJ375A	Microsoft Edge WebView2 Runtime	148.0.3967.70	Improper access control in Micro...	High	CVE-2026-58525
DESKTOP-2GJ375A	Microsoft Edge WebView2 Runtime	148.0.3967.70	Integer overflow or wraparound L...	High	CVE-2026-57974
DESKTOP-2GJ375A	Microsoft Edge WebView2 Runtime	148.0.3967.70	Heap-based buffer overflow in M...	High	CVE-2026-56645
DESKTOP-2GJ375A	Microsoft Edge WebView2 Runtime	148.0.3967.70	User interface (ui) misrepresenta...	Medium	CVE-2026-45488

Rows per page: 15

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

36°C Mostly sunny

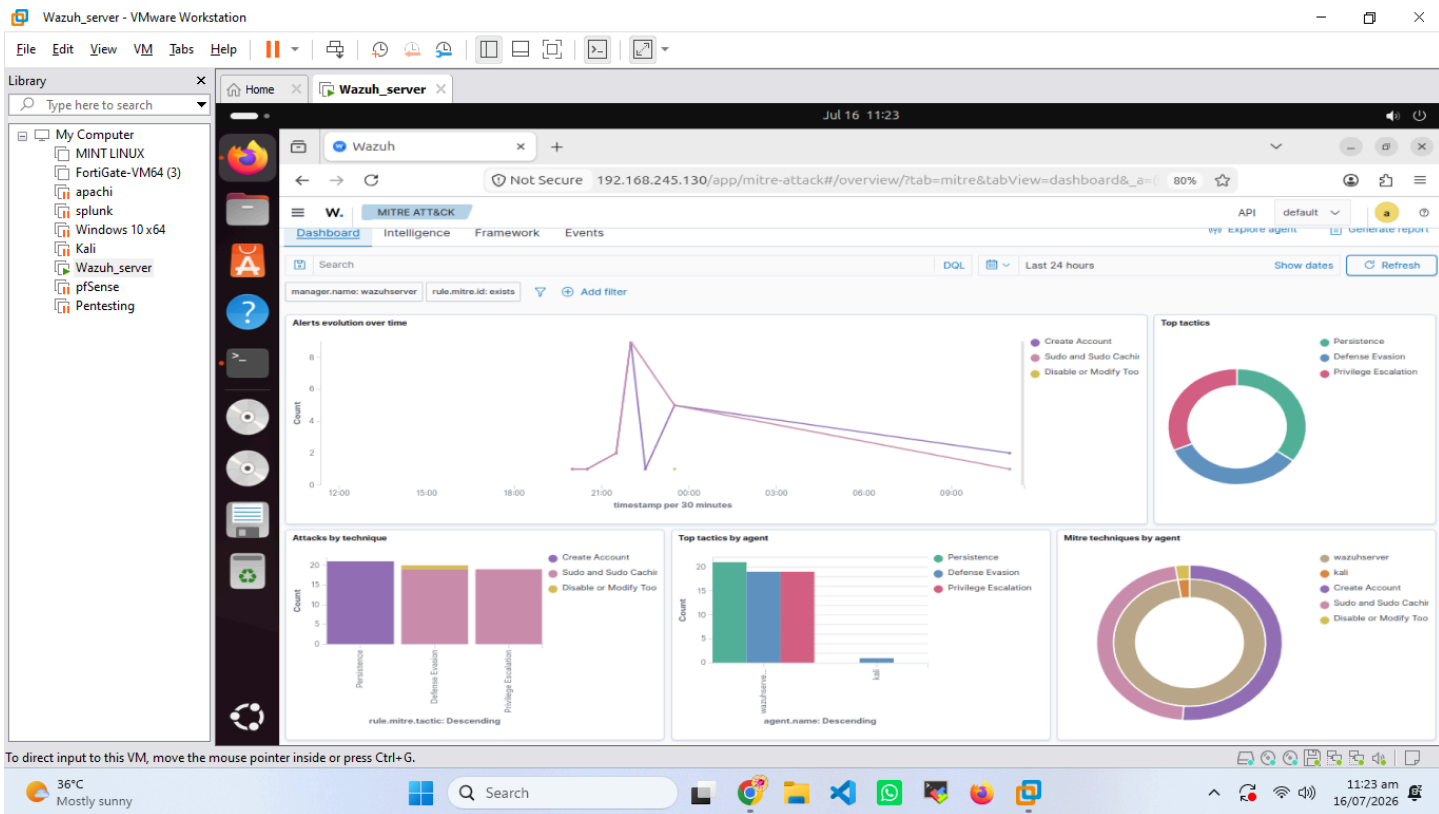
Search

11:29 am 16/07/2026

MITRE ATT&CK Mapping Implementation

Wazuh's built-in MITRE ATT&CK framework was utilized to map detected events to corresponding adversarial tactics and techniques. Alerts from Suricata, pfSense, and endpoint agents were categorized under tactics such as **Initial Access**, **Execution**, and **Command and Control**.

This mapping provided contextual understanding of attack progression and improved incident triage efficiency.



Wazuh_server - VMware Workstation

File Edit View VM Tabs Help

Library

My Computer

- MINT LINUX
- FortiGate-VM64 (3)
- apachi
- splunk
- Windows 10 x64
- Kali
- Wazuh_server
- pfSense
- Pentesting

Wazuh

Jul 16 11:22

Not Secure 192.168.245.130/app/mitre-attack#/overview/?tab=mitre&tabView=events&a=(filter) 80%

MITRE ATTACK

Export Formatted Reset view 739 available fields Columns Density 1 fields sorted Full screen

41 hits

Jul 15, 2026 @ 11:21:51.296 - Jul 16, 2026 @ 11:21:51.296

timestamp	agent.name	rule.mitre.id	rule.mitre.tactic	rule.description	rule.level	rule.id
Jul 16, 2026 @ 11:15:14.602	wazuhsrver	T1136	Persistence	New user account created on the system - Investi...	10	100001
Jul 16, 2026 @ 11:14:16.548	wazuhsrver	T1136	Persistence	New user account created on the system - Investi...	10	100001
Jul 16, 2026 @ 11:14:16.547	wazuhsrver	T1548.003	Privilege Escalation, Defense Evasion	Successful sudo to ROOT executed.	3	5402
Jul 15, 2026 @ 23:48:55.582	wazuhsrver	T1136	Persistence	New user account created on the system - Investi...	10	100001
Jul 15, 2026 @ 23:48:55.582	wazuhsrver	T1548.003	Privilege Escalation, Defense Evasion	Successful sudo to ROOT executed.	3	5402
Jul 15, 2026 @ 23:46:59.444	wazuhsrver	T1136	Persistence	New user account created on the system - Investi...	10	100001
Jul 15, 2026 @ 23:46:59.444	wazuhsrver	T1548.003	Privilege Escalation, Defense Evasion	Successful sudo to ROOT executed.	3	5402
Jul 15, 2026 @ 23:46:53.442	wazuhsrver	T1136	Persistence	New user account created on the system - Investi...	10	100001
Jul 15, 2026 @ 23:46:53.442	wazuhsrver	T1548.003	Privilege Escalation, Defense Evasion	Successful sudo to ROOT executed.	3	5402
Jul 15, 2026 @ 23:46:09.426	wazuhsrver	T1136	Persistence	New user account created on the system - Investi...	10	100001
Jul 15, 2026 @ 23:46:09.425	wazuhsrver	T1548.003	Privilege Escalation, Defense Evasion	Successful sudo to ROOT executed.	3	5402
Jul 15, 2026 @ 23:45:53.418	wazuhsrver	T1136	Persistence	New user account created on the system - Investi...	10	100001
Jul 15, 2026 @ 23:45:53.418	wazuhsrver	T1548.003	Privilege Escalation, Defense Evasion	Successful sudo to ROOT executed.	3	5402
Jul 15, 2026 @ 23:34:34.384	kali	T1562.001	Defense Evasion	Wazuh agent disconnected.	3	504
Jul 15, 2026 @ 22:49:50.902	wazuhsrver	T1136	Persistence	New user account created on the system - Investi...	10	100001

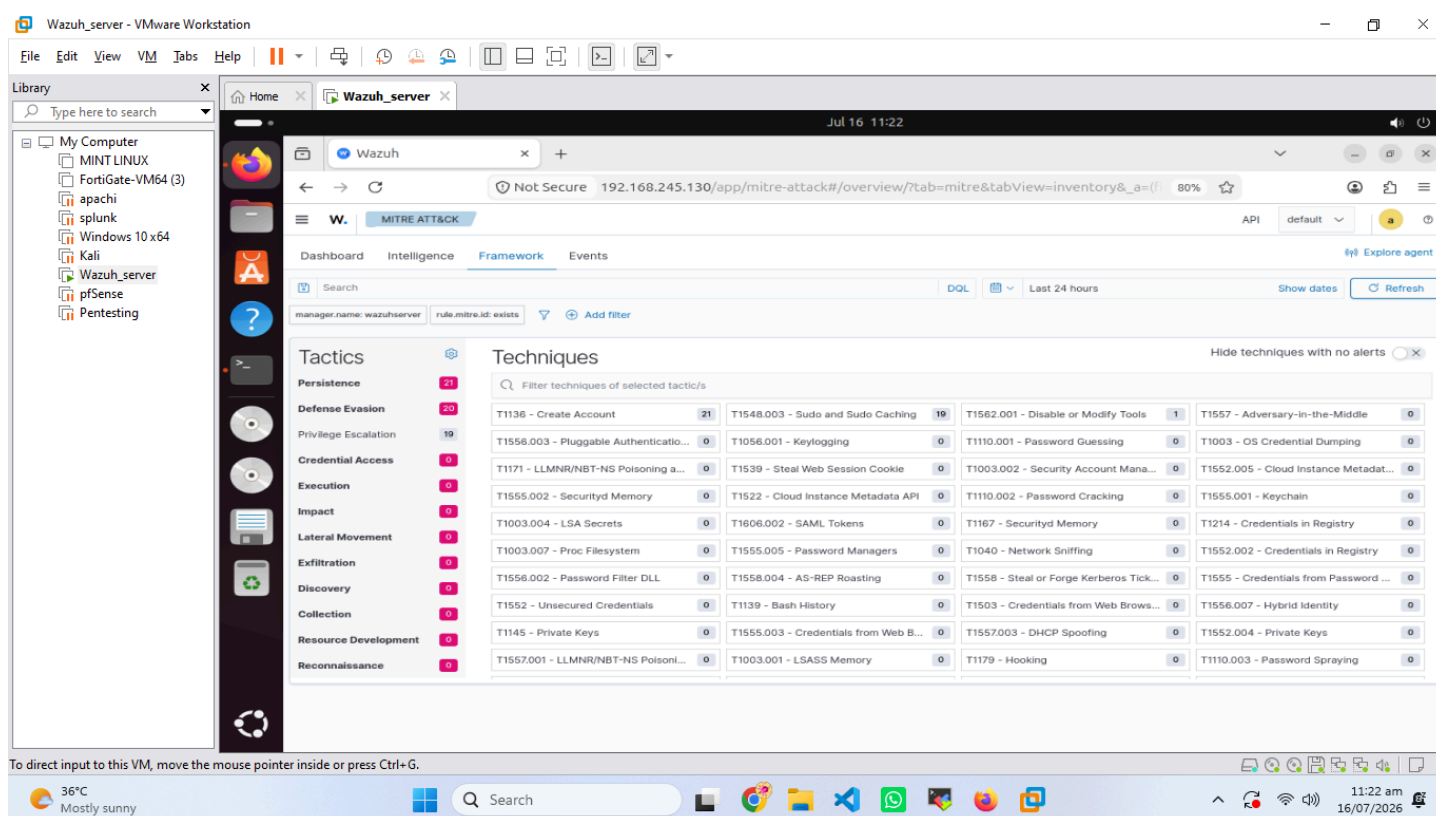
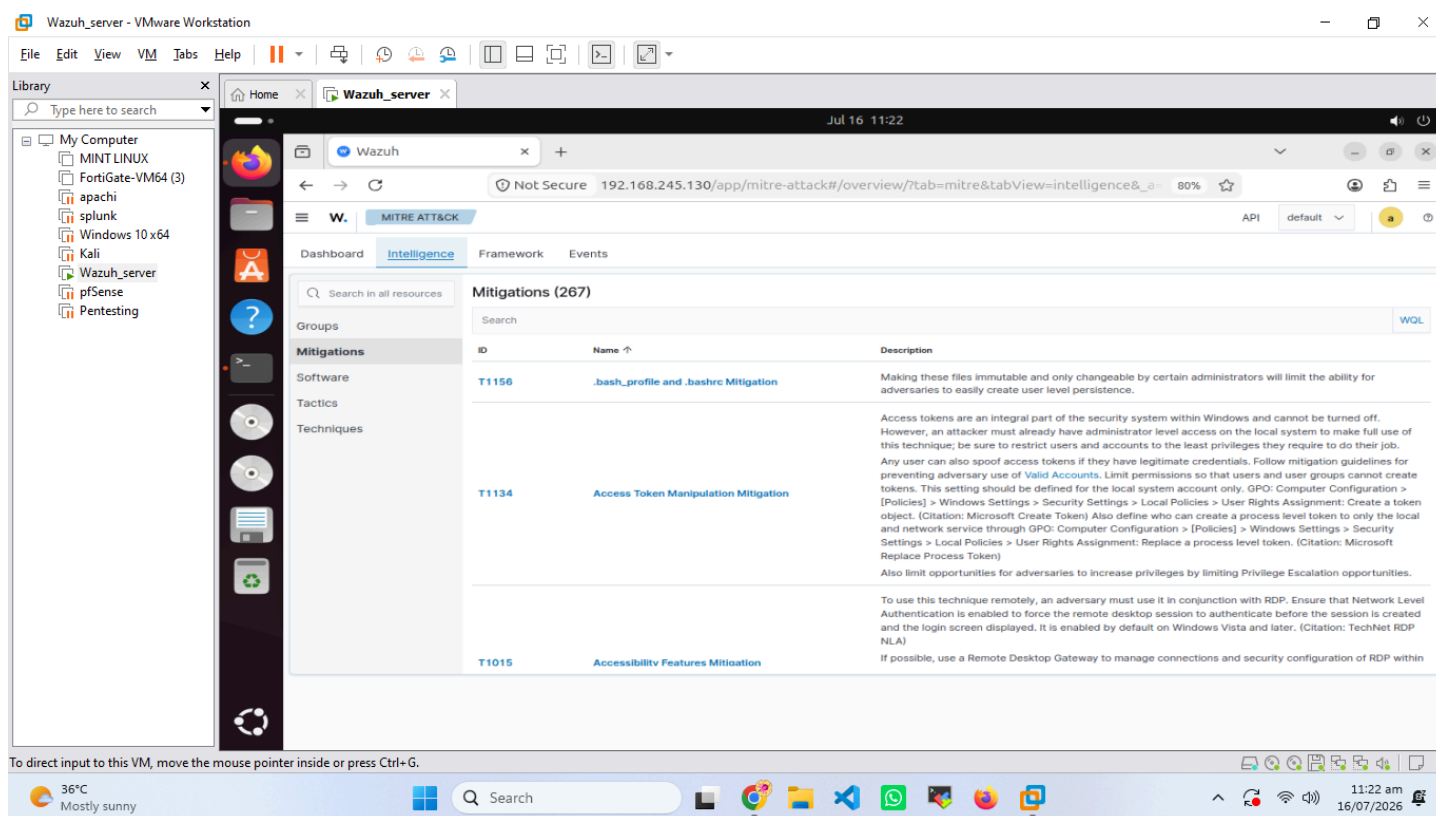
Rows per page: 15

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

36°C Mostly sunny

Search

11:22 am 16/07/2026



Conclusion

Custom SOC Dashboard Creation

A customized **Wazuh Dashboard** was developed to visualize key SOC metrics, including:

- Firewall traffic and blocked connections
- IDS alerts from Suricata

- Endpoint security events
- VirusTotal threat intelligence results
- Vulnerability statistics
- MITRE ATT&CK tactic distribution

This dashboard provided a single-pane-of-glass view for SOC analysts to monitor the entire defensive ecosystem.

Observations

- pfSense effectively centralized network control and improved visibility.
- Integration with Wazuh enabled real-time correlation of network and endpoint events.
- VirusTotal enrichment added valuable context to threat detections.
- MITRE ATT&CK mapping enhanced analytical depth for incident response.
- The SOC dashboard streamlined monitoring and reporting processes.

Challenges Faced

- Initial misconfiguration of syslog forwarding caused incomplete log ingestion.
- NAT and routing conflicts required reassigning network adapters in VMware.
- VirusTotal API rate limits occasionally delayed enrichment results.
- Fine-tuning Wazuh decoders for pfSense logs required iterative testing.

Lessons Learned

- Proper network segmentation is essential for scalable SOC operations.
- Centralized logging simplifies correlation and forensic analysis.
- Threat intelligence integration significantly improves detection fidelity.
- Continuous tuning of SIEM rules and dashboards enhances operational efficiency.

Conclusion

Week 4 marked the completion of **Phase One – SOC Operations and SIEM Foundations**. The deployment of pfSense, integration of threat intelligence, vulnerability detection, and dashboard customization transformed the lab into a fully functional SOC environment. The infrastructure now supports comprehensive monitoring, detection, and analysis capabilities, laying the groundwork for advanced incident response and threat hunting in subsequent phases.